

PowerShell Commands and Outputs

Windows PowerShell

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```
PS C:\Users\Manisha Rajalingam> mongosh --version
```

2.12.0

```
PS C:\Users\Manisha Rajalingam> mongosh
```

Current Mongosh Log ID: 6ac09789b1c42f9b210034dd

Connecting to:

mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0

Using MongoDB: 8.3.2

Using Mongosh: 2.12.0

For mongosh info see: <https://www.mongodb.com/docs/mongosh-shell/>

To help improve our products, pseudo-anonymous usage data is collected from interactive and agentic sessions only, and sent to MongoDB periodically (<https://www.mongodb.com/legal/privacy-policy>).

You can opt-out by running the `disableTelemetry()` command.

The server generated these startup warnings when booting

2026-09-19T06:18:49.564+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

```
test> show dbs
```

admin	40.00 KiB
-------	-----------

ai_roadmap_db	104.00 KiB
---------------	------------

careerRoadmap	96.00 KiB
---------------	-----------

```
config          36.00 KiB
local           80.00 KiB
placement_portal 112.00 KiB

test> use collegeDB
switched to db collegeDB

collegeDB> db
collegeDB

collegeDB> db.createCollection("students")
{ ok: 1 }

collegeDB> show collections
students

collegeDB> db.students.insertOne({
|  rollno: 101,
|  name: "Arun",
|  department: "CSE",
|  year: 3,
|  marks: 85
| })
{
  acknowledged: true,
  insertedId: ObjectId('6ac0981cb1c42f9b210034de')
}

collegeDB> db.students.insertMany([
|  { rollno: 102, name: "Priya", department: "ECE", year: 3, marks: 92 },
|  { rollno: 103, name: "Karthik", department: "CSE", year: 2, marks: 78 },
|  { rollno: 104, name: "Meena", department: "EEE", year: 3, marks: 88 },
|  { rollno: 105, name: "Rahul", department: "CSE", year: 4, marks: 67 }
| ])
{
  acknowledged: true,
```

```
insertedIds: {
  '0': ObjectId('6ac09833b1c42f9b210034df'),
  '1': ObjectId('6ac09833b1c42f9b210034e0'),
  '2': ObjectId('6ac09833b1c42f9b210034e1'),
  '3': ObjectId('6ac09833b1c42f9b210034e2')
}
}
```

```
collegeDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac0981cb1c42f9b210034de'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034df'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e0'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
```

```

    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e1'),
    rollno: 104,
    name: 'Meena',
    department: 'EEE',
    year: 3,
    marks: 88
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e2'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
  }
]
collegeDB> db.students.insertMany([
| { rollno: 106, name: "Divya", department: "ECE", year: 2, marks: 81, city: "Chennai" },
| { rollno: 107, name: "Sanjay", department: "CSE", year: 3, marks: 95, city: "Coimbatore"
| },
| { rollno: 108, name: "Anitha", department: "EEE", year: 4, marks: 73, city: "Salem" },
| { rollno: 109, name: "Vijay", department: "CSE", year: 3, marks: 89, city: "Madurai" },
| { rollno: 110, name: "Nithya", department: "ECE", year: 3, marks: 84, city: "Trichy" }
| ])
{
  acknowledged: true,
  insertedIds: {

```

```
'0': ObjectId('6ac0a10eb1c42f9b210034e3'),
'1': ObjectId('6ac0a10eb1c42f9b210034e4'),
'2': ObjectId('6ac0a10eb1c42f9b210034e5'),
'3': ObjectId('6ac0a10eb1c42f9b210034e6'),
'4': ObjectId('6ac0a10eb1c42f9b210034e7')
}
}
```

```
collegeDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac0981cb1c42f9b210034de'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034df'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e0'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
```

```
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e1'),
    rollno: 104,
    name: 'Meena',
    department: 'EEE',
    year: 3,
    marks: 88
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e2'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e3'),
    rollno: 106,
    name: 'Divya',
    department: 'ECE',
    year: 2,
    marks: 81,
    city: 'Chennai'
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e4'),
```

```
rollno: 107,  
name: 'Sanjay',  
department: 'CSE',  
year: 3,  
marks: 95,  
city: 'Coimbatore'  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e5'),  
  rollno: 108,  
  name: 'Anitha',  
  department: 'EEE',  
  year: 4,  
  marks: 73,  
  city: 'Salem'  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e6'),  
  rollno: 109,  
  name: 'Vijay',  
  department: 'CSE',  
  year: 3,  
  marks: 89,  
  city: 'Madurai'  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e7'),  
  rollno: 110,  
  name: 'Nithya',  
  department: 'ECE',
```

```
    year: 3,
    marks: 84,
    city: 'Trichy'
  }
]
collegeDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('6ac0981cb1c42f9b210034de'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e0'),
    rollno: 103,
    name: 'Karthik',
    department: 'CSE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e2'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
  }
]
```

```
    },  
    {  
      _id: ObjectId('6ac0a10eb1c42f9b210034e4'),  
      rollno: 107,  
      name: 'Sanjay',  
      department: 'CSE',  
      year: 3,  
      marks: 95,  
      city: 'Coimbatore'  
    },  
    {  
      _id: ObjectId('6ac0a10eb1c42f9b210034e6'),  
      rollno: 109,  
      name: 'Vijay',  
      department: 'CSE',  
      year: 3,  
      marks: 89,  
      city: 'Madurai'  
    }  
  ]  
collegeDB> db.students.find({ marks: { $gt: 80 } })  
[  
  {  
    _id: ObjectId('6ac0981cb1c42f9b210034de'),  
    rollno: 101,  
    name: 'Arun',  
    department: 'CSE',  
    year: 3,  
    marks: 85  
  },  
]
```

```
{
  _id: ObjectId('6ac09833b1c42f9b210034df'),
  rollno: 102,
  name: 'Priya',
  department: 'ECE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e1'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e3'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e4'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
```

```
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e6'),
    rollno: 109,
    name: 'Vijay',
    department: 'CSE',
    year: 3,
    marks: 89,
    city: 'Madurai'
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e7'),
    rollno: 110,
    name: 'Nithya',
    department: 'ECE',
    year: 3,
    marks: 84,
    city: 'Trichy'
  }
]
collegeDB> db.students.find({}, { name: 1, marks: 1, _id: 0 })
[
  { name: 'Arun', marks: 85 },
  { name: 'Priya', marks: 92 },
  { name: 'Karthik', marks: 78 },
  { name: 'Meena', marks: 88 },
  { name: 'Rahul', marks: 67 },
```

```
{ name: 'Divya', marks: 81 },  
{ name: 'Sanjay', marks: 95 },  
{ name: 'Anitha', marks: 73 },  
{ name: 'Vijay', marks: 89 },  
{ name: 'Nithya', marks: 84 }  
]
```

```
collegeDB> db.students.find({ department: "CSE", year: 3 })
```

```
[  
  {  
    _id: ObjectId('6ac0981cb1c42f9b210034de'),  
    rollno: 101,  
    name: 'Arun',  
    department: 'CSE',  
    year: 3,  
    marks: 85  
  },  
  {  
    _id: ObjectId('6ac0a10eb1c42f9b210034e4'),  
    rollno: 107,  
    name: 'Sanjay',  
    department: 'CSE',  
    year: 3,  
    marks: 95,  
    city: 'Coimbatore'  
  },  
  {  
    _id: ObjectId('6ac0a10eb1c42f9b210034e6'),  
    rollno: 109,  
    name: 'Vijay',  
    department: 'CSE',
```

```
    year: 3,
    marks: 89,
    city: 'Madurai'
  }
]
collegeDB> db.students.find({
| $or: [
|   { department: "CSE" },
|   { department: "ECE" }
| ]
| })
[
  {
    _id: ObjectId('6ac0981cb1c42f9b210034de'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034df'),
    rollno: 102,
    name: 'Priya',
    department: 'ECE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034e0'),
```

```
rollno: 103,  
name: 'Karthik',  
department: 'CSE',  
year: 2,  
marks: 78  
,  
{  
  _id: ObjectId('6ac09833b1c42f9b210034e2'),  
  rollno: 105,  
  name: 'Rahul',  
  department: 'CSE',  
  year: 4,  
  marks: 67  
,  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e3'),  
  rollno: 106,  
  name: 'Divya',  
  department: 'ECE',  
  year: 2,  
  marks: 81,  
  city: 'Chennai'  
,  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e4'),  
  rollno: 107,  
  name: 'Sanjay',  
  department: 'CSE',  
  year: 3,  
  marks: 95,
```

```
    city: 'Coimbatore'
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e6'),
    rollno: 109,
    name: 'Vijay',
    department: 'CSE',
    year: 3,
    marks: 89,
    city: 'Madurai'
  },
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e7'),
    rollno: 110,
    name: 'Nithya',
    department: 'ECE',
    year: 3,
    marks: 84,
    city: 'Trichy'
  }
]
collegeDB> db.students.find().sort({ marks: 1 })
[
  {
    _id: ObjectId('6ac09833b1c42f9b210034e2'),
    rollno: 105,
    name: 'Rahul',
    department: 'CSE',
    year: 4,
    marks: 67
```

```
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e5'),  
  rollno: 108,  
  name: 'Anitha',  
  department: 'EEE',  
  year: 4,  
  marks: 73,  
  city: 'Salem'  
},  
{  
  _id: ObjectId('6ac09833b1c42f9b210034e0'),  
  rollno: 103,  
  name: 'Karthik',  
  department: 'CSE',  
  year: 2,  
  marks: 78  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e3'),  
  rollno: 106,  
  name: 'Divya',  
  department: 'ECE',  
  year: 2,  
  marks: 81,  
  city: 'Chennai'  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e7'),  
  rollno: 110,
```

```
name: 'Nithya',
department: 'ECE',
year: 3,
marks: 84,
city: 'Trichy'
},
{
  _id: ObjectId('6ac0981cb1c42f9b210034de'),
  rollno: 101,
  name: 'Arun',
  department: 'CSE',
  year: 3,
  marks: 85
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e1'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e6'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
```

```
    },  
    {  
      _id: ObjectId('6ac09833b1c42f9b210034df'),  
      rollno: 102,  
      name: 'Priya',  
      department: 'ECE',  
      year: 3,  
      marks: 92  
    },  
    {  
      _id: ObjectId('6ac0a10eb1c42f9b210034e4'),  
      rollno: 107,  
      name: 'Sanjay',  
      department: 'CSE',  
      year: 3,  
      marks: 95,  
      city: 'Coimbatore'  
    }  
  ]  
collegeDB> db.students.find().sort({ marks: -1 })  
[  
  {  
    _id: ObjectId('6ac0a10eb1c42f9b210034e4'),  
    rollno: 107,  
    name: 'Sanjay',  
    department: 'CSE',  
    year: 3,  
    marks: 95,  
    city: 'Coimbatore'  
  },  
  {  
    _id: ObjectId('6ac09833b1c42f9b210034df'),  
    rollno: 102,  
    name: 'Priya',  
    department: 'ECE',  
    year: 3,  
    marks: 92  
  }  
]
```

```
{
  _id: ObjectId('6ac09833b1c42f9b210034df'),
  rollno: 102,
  name: 'Priya',
  department: 'ECE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e6'),
  rollno: 109,
  name: 'Vijay',
  department: 'CSE',
  year: 3,
  marks: 89,
  city: 'Madurai'
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e1'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
  year: 3,
  marks: 88
},
{
  _id: ObjectId('6ac0981cb1c42f9b210034de'),
  rollno: 101,
  name: 'Arun',
  department: 'CSE',
```

```
year: 3,
marks: 85
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e7'),
  rollno: 110,
  name: 'Nithya',
  department: 'ECE',
  year: 3,
  marks: 84,
  city: 'Trichy'
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e3'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e0'),
  rollno: 103,
  name: 'Karthik',
  department: 'CSE',
  year: 2,
  marks: 78
},
{
```

```
_id: ObjectId('6ac0a10eb1c42f9b210034e5'),
rollno: 108,
name: 'Anitha',
department: 'EEE',
year: 4,
marks: 73,
city: 'Salem'
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e2'),
  rollno: 105,
  name: 'Rahul',
  department: 'CSE',
  year: 4,
  marks: 67
}
]
collegeDB> db.students.find().sort({ marks: -1 }).limit(3)
[
  {
    _id: ObjectId('6ac0a10eb1c42f9b210034e4'),
    rollno: 107,
    name: 'Sanjay',
    department: 'CSE',
    year: 3,
    marks: 95,
    city: 'Coimbatore'
  },
  {
    _id: ObjectId('6ac09833b1c42f9b210034df'),
```

```
rollno: 102,  
name: 'Priya',  
department: 'ECE',  
year: 3,  
marks: 92  
},  
{  
  _id: ObjectId('6ac0a10eb1c42f9b210034e6'),  
rollno: 109,  
name: 'Vijay',  
department: 'CSE',  
year: 3,  
marks: 89,  
city: 'Madurai'  
}  
]  
collegeDB> db.students.countDocuments({ department: "CSE" })  
5  
collegeDB> db.students.updateOne(  
| { rollno: 101 },  
| { $set: { marks: 90 } }  
| )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
collegeDB> db.students.updateMany(
```

```

| { department: "CSE" },
| { $set: { status: "Active" } }
|)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
collegeDB> db.students.deleteOne({ rollno: 105 })
{ acknowledged: true, deletedCount: 1 }
collegeDB> db.students.distinct("department")
[ 'CSE', 'ECE', 'EEE' ]
collegeDB> db.students.aggregate([
| {
|   $group: {
|     _id: "$department",
|     averageMarks: { $avg: "$marks" }
|   }
| }
|])
[
  { _id: 'EEE', averageMarks: 80.5 },
  { _id: 'CSE', averageMarks: 88 },
  { _id: 'ECE', averageMarks: 85.66666666666667 }
]
collegeDB> db.students.find()
[
  {

```

```
_id: ObjectId('6ac0981cb1c42f9b210034de'),
rollno: 101,
name: 'Arun',
department: 'CSE',
year: 3,
marks: 90,
status: 'Active'
},
{
  _id: ObjectId('6ac09833b1c42f9b210034df'),
  rollno: 102,
  name: 'Priya',
  department: 'ECE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e0'),
  rollno: 103,
  name: 'Karthik',
  department: 'CSE',
  year: 2,
  marks: 78,
  status: 'Active'
},
{
  _id: ObjectId('6ac09833b1c42f9b210034e1'),
  rollno: 104,
  name: 'Meena',
  department: 'EEE',
```

```
year: 3,
marks: 88
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e3'),
  rollno: 106,
  name: 'Divya',
  department: 'ECE',
  year: 2,
  marks: 81,
  city: 'Chennai'
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e4'),
  rollno: 107,
  name: 'Sanjay',
  department: 'CSE',
  year: 3,
  marks: 95,
  city: 'Coimbatore',
  status: 'Active'
},
{
  _id: ObjectId('6ac0a10eb1c42f9b210034e5'),
  rollno: 108,
  name: 'Anitha',
  department: 'EEE',
  year: 4,
  marks: 73,
  city: 'Salem'
```

```
    },  
    {  
      _id: ObjectId('6ac0a10eb1c42f9b210034e6'),  
      rollno: 109,  
      name: 'Vijay',  
      department: 'CSE',  
      year: 3,  
      marks: 89,  
      city: 'Madurai',  
      status: 'Active'  
    },  
    {  
      _id: ObjectId('6ac0a10eb1c42f9b210034e7'),  
      rollno: 110,  
      name: 'Nithya',  
      department: 'ECE',  
      year: 3,  
      marks: 84,  
      city: 'Trichy'  
    }  
  ]  
collegeDB>
```